Stochastics of News Article Propagation: Joint Modeling of Counts and Reliability Composition in Event-Driven News Cascades

Harry Wang Sriyan Madugula Ruthesh Thavamani

MATH 460 — Stochastic Processes

Abstract

We study the news cascade triggered by a single high-impact event as a pair of coupled stochastic processes: the daily article count $N(t)$ and the reliable-vs-unreliable composition $\pi_{\text{pos}}(t)$. Using the 2017 Las Vegas shooting as a primary case study (CC-News + NewsGuard, 91 days, 2,702 articles), we fit and compare four counting-process models for $N(t)$ and four distributional models for $\pi_{\text{pos}}(t)$ on a common likelihood scale. For counts, a hybrid Inhomogeneous-Poisson plus Hawkes model wins decisively on AIC, BIC, and log-likelihood, as the IHP accounts for the initial exogenous shock, while the Hawkes accounts for the subsequent self-excitation. A profile-likelihood sweep of the Hawkes decay rate collapses the optimal Hawkes kernel to an AR(1) one-step lag $\lambda(t) = \mu + nN(t-1)$. This Markovian structure on the counts directly motivates the Part-B reliability models: we treat the reliability label as a first-order Markov chain on a binary state space and test homogeneous, regime-switching, and mean-field parameterizations against a bivariate Hawkes baseline model using a shared conditional binomial likelihood. The results show that the bivariate Hawkes model and the non-homogeneous Markov models are nearly indistinguishable on fit, as they exhaust the signal and start fitting to the noise. A brief replication on the 2017 Hurricane Harvey data shows that the same methodology applies but produces different results due to Hurricane Harvey's gradual endogenous build-up.

1 Introduction

A single high-impact event causes a cascade of news articles. Two observable quantities of that cascade are simultaneously stochastic: *how much* is published on each day after the event ($N(t)$) and *how reliable* that coverage is ($\pi_{\text{pos}}(t)$, the fraction of articles published by reliable outlets). Most existing work treats one of these in isolation. We argue that both deserve a joint stochastic-process treatment on the same event timeline, and that doing so is feasible with public data.

Research questions.

- **Part A.** What stochastic process best fits the daily article count $N(t)$ following an event?
- **Part B.** What process governs the reliable-vs-unreliable composition $\pi_{\text{pos}}(t)$ over the same window?

Running example. The 2017 Las Vegas shooting is our primary case study. The event has a singular, exogenous, well-defined origin (no prior coverage ramp-up), a sharp day-0 spike (489 articles), and a long, decaying tail (91 days). Reliable and unreliable outlets both engaged with the story so the reliability composition is non-trivial. However, it should be noted that the there are a majority of reliable outlets (63.5%), which could result in some model biases. Hurricane Harvey 2017 serves as a secondary cross-event replication.

Prior work. Three threads in the literature are relevant.

Point-process models of online attention. Crane and Sornette [1] used Hawkes-type response functions to separate endogenous from exogenous bursts in YouTube views, which is conceptually similar to our Part-A hybrid. Zhao et al. (SEISMIC, [2]) and Mishra, Rizoiu, and Xie [3] extended self-exciting processes to tweet/retweet cascades. Daley and Vere-Jones [4] provide the canonical reference for inhomogeneous Poisson and Hawkes processes on point patterns.

Misinformation and reliability. Vosoughi, Roy, and Aral [5] showed descriptively that false news travels faster and farther on Twitter than true news, but did not write down a generative stochastic model. Allcott and Gentzkow [6] measured the economics of fake news during the 2016 U.S. election. Tambuscio et al. [7] used SIR-style compartmental models for hoax diffusion. This is closest in spirit to our Markov-chain work, but in the context of continuous-time epidemiology rather than discrete-time time-series.

Reliability scoring infrastructure. NewsGuard and Media Bias / Fact Check provide outlet-level reliability scores on a scale from 0 to 100. To convert these scores to a binary label, we threshold at the midpoint so that scores above 50 are considered reliable and scores below 50 are considered unreliable. These scores are typically used as features for downstream classification, rarely as the response variable in a stochastic time-series model. We use NewsGuard explicitly as the response.

Gap and contributions. We are not aware of prior work that provides a joint, event-level, stochastic-process treatment of *both* the count and the reliability composition on the *same* timeline, fit with the *same* statistical scoring. Concretely, this paper contributes:

1. A hybrid IHP + Hawkes model for the count process that decomposes a news cascade into an exogenous trigger and endogenous amplification operating at two distinct timescales.

2. An empirical bridge from Part A to Part B: the Hawkes profile likelihood collapses to an AR(1) limit, motivating Markov-chain modeling of the reliability state. This narrative thread is the backbone of §2.3.

3. A shared conditional binomial likelihood that puts a multivariate Hawkes process and a family of Markov-chain models on a single AIC/BIC scale, despite having native likelihoods of different types. This allows us to directly compare how these models perform on the same task of predicting the reliability composition.

2 Methods

2.1 Data and preprocessing

We use the news subset of the Common Crawl (CC-News), which provides URL, publication date, and article body text. We left-join against the public NewsGuard domain-level reliability dataset on the article's URL domain (see `loader.py`), so that each article inherits the score of its publishing outlet. The continuous NewsGuard score is thresholded at the median to yield a binary reliability label $\{-1, +1\}$ (unreliable / reliable). Event-relevant articles are selected by a keyword filter (per-event TOML configuration); missing dates are zero-filled so every model sees the same regular daily grid.

For the Las Vegas 2017 main analysis the window is 2017-10-02 to 2017-12-31 ($T = 91$ days, zero-filled), with 2,702 articles total ($N_{\text{neg}} = 986$, $N_{\text{pos}} = 1{,}716$; empirical split 0.365/0.635), a peak of 489 articles on day 0, and 5 days with zero articles (skipped in the binomial likelihood).

2.2 Part A: counting-process models for $N(t)$

We fit three baseline counting-process models and a four-parameter hybrid. All four models are scored on the same conditional Poisson log-likelihood including the $\log N!$ normalization,

$$\ell = \sum_{t=0}^{T-1} \Big[N(t) \log \lambda(t) - \lambda(t) - \log N(t)! \Big], \tag{1}$$

so that AIC and BIC are directly comparable across families.

Standard Poisson (SP, 1 parameter). $\lambda(t) = \lambda$ for all t; MLE in closed form, $\hat{\lambda} = \overline{N}$. Included as a null baseline.

Inhomogeneous Poisson (IHP, 3 parameters). $\lambda(t) = A\, e^{-\beta t} + c$ with $A \geq 0$ the initial amplitude above baseline, $\beta > 0$ the exponential decay rate, and $c \geq 0$ the long-run baseline. Each day's count is an independent Poisson draw whose rate decays deterministically with absolute calendar time: this captures the "news-cycle decay" phenomenon. Optimized by Nelder–Mead with multi-start.

Discrete-time Hawkes (2 parameters in the AR-1 limit, 3 in general). We use the recursive form

$$\lambda(t) = \mu + \alpha\, R(t), \qquad R(t) = e^{-\beta} \big[R(t-1) + N(t-1) \big], \quad R(0) = 0, \tag{2}$$

with the stationary reparameterization

$$n = \frac{\alpha}{e^{\beta} - 1} \in (0, 1), \tag{3}$$

where n is the *branching ratio* (the expected number of secondary articles triggered per article). $\lambda(t)$ is conditional on the past counts $N(0), \ldots, N(t-1)$, so the log-likelihood (1) must be evaluated recursively rather than with a close-form MLE on independent Poisson counts.

Hybrid IHP + Hawkes (AR-1, 4 parameters). The hybrid combines exogenous decay with endogenous self-excitation:

$$\lambda(0) = A + c, \qquad \lambda(t) = A\, e^{-\beta_0 t} + c + n\, N(t-1), \quad t \geq 1. \tag{4}$$

The $A\, e^{-\beta_0 t} + c$ term captures exogenous news-cycle decay, while the $n\, N(t-1)$ term captures endogenous next-day amplification. Hybrid-AR1 strictly nests both pure IHP (set $n = 0$) and pure Hawkes-AR1 (drop the IHP term and rename $c \to \mu$), enabling likelihood-ratio tests in addition to AIC/BIC.

Optimization. All models are fit with Nelder–Mead from a multi-start grid of initializations to dodge local minima. Stationarity of the Hawkes process is checked at the fitted value; for the multivariate Hawkes of §2.4 stationarity is enforced as a spectral-radius constraint on the branching matrix.

2.3 The AR(1) $\to$ Markov bridge between Part A and Part B

A diagnostic on the Part-A Hawkes that profoundly shapes Part B is the *profile log-likelihood of the kernel decay* β. Fixing β on a log-grid from 0.05 to 32 and re-optimizing (μ, n), the profile log-likelihood is monotone increasing in β and plateaus for $\beta \geq 10$ (Table 1). In the limit $\beta \to \infty$ the exponential kernel $e^{-\beta\Delta}$ collapses to a delta at lag $\Delta = 1$, and the Hawkes intensity reduces to

$$\lambda(t) \approx \mu + n\,N(t-1). \tag{5}$$

This is a discrete *AR(1) model* for the intensity of the count process: the next-day rate depends only on yesterday's count, which is exactly the conditional independence structure of a first-order Markov chain.

This empirical AR(1) collapse motivates the Part-B model family. For the binary reliability state, the natural generalization of (5) is a first-order Markov chain on the state space $\{\text{neg}, \text{pos}\}$, parameterized by a 2×2 transition matrix P that may be *homogeneous* (a single P), *regime-switching* (two $P^{(\cdot)}$ matrices indexed by yesterday's majority state), or *mean-field* (the entries of P_t depend smoothly on yesterday's empirical proportion $\pi_{\text{pos}}(t-1)$). This thread is summarized in Figure 1.

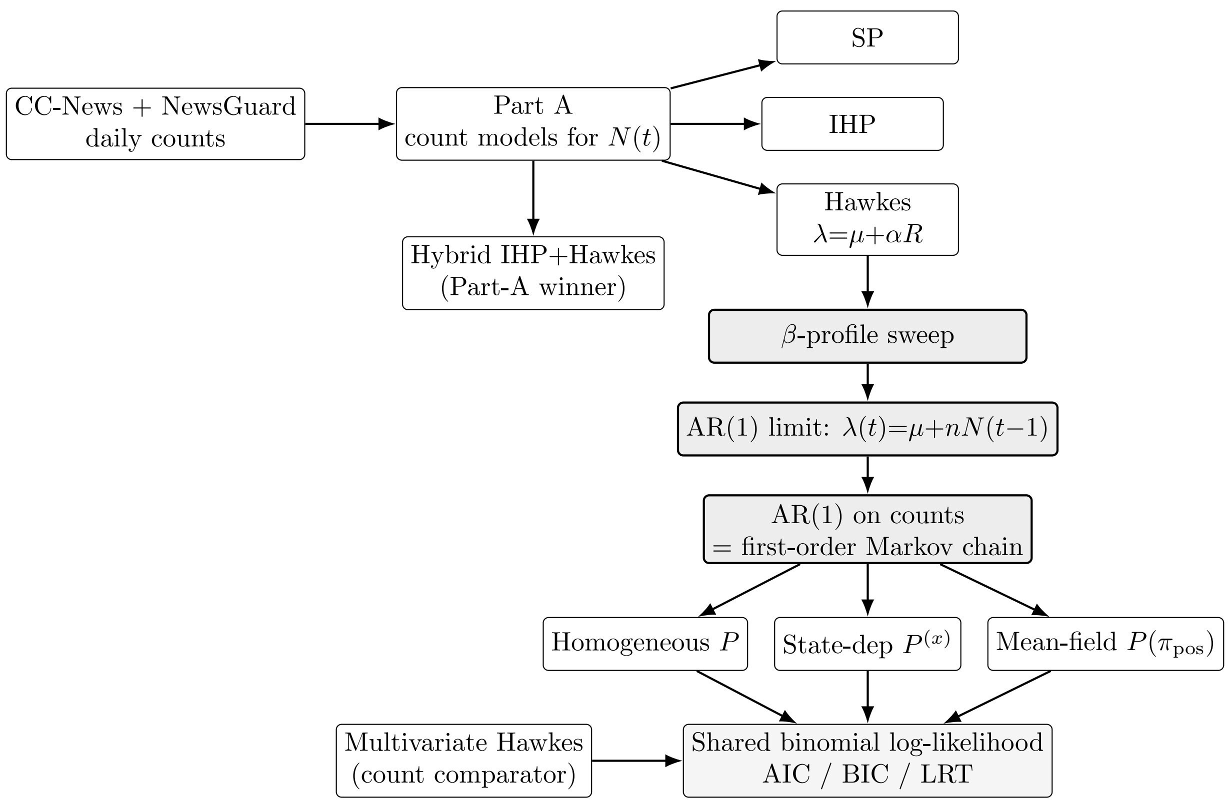

Figure 1: The narrative thread connecting Part A and Part B. The Hawkes β-profile sweep on daily-resolution counts plateaus in the AR(1) limit, and an AR(1) on a discrete state is precisely a first-order Markov chain. This motivates a Markov-chain family for the binary reliability state in Part B, scored against the multivariate-Hawkes count model on a common conditional binomial likelihood.

2.4 Part B: distributional models for the reliability split

Part B asks: given the daily total $N_{\text{tot}}(t) = N_{\text{neg}}(t) + N_{\text{pos}}(t)$, what process explains the daily *reliable fraction* $\pi_{\text{pos}}(t) = N_{\text{pos}}(t)/N_{\text{tot}}(t)$?

Shared conditional binomial likelihood. Hawkes natively scores the pair of counts $(N_{\text{neg}}(t), N_{\text{pos}}(t))$ under a multivariate Poisson log-likelihood; a Markov chain natively scores a one-step transition under a multinomial / binomial on the proportion. The two native AICs live on different scales, so direct comparison is meaningless. To put all candidates on one scale we use the Poisson–binomial superposition identity: if $X \sim \text{Pois}(\lambda_1)$ and $Y \sim \text{Pois}(\lambda_2)$ independently, then $X \mid X + Y = n \sim \text{Binomial}(n, \lambda_1/(\lambda_1 + \lambda_2))$. Applied to our setup,

$$N_{\text{pos}}(t) \,\big|\, N_{\text{tot}}(t) \;\sim\; \text{Binomial}\big(N_{\text{tot}}(t),\, q_t\big), \tag{6}$$

with model-specific q_t. The per-day log-likelihood is

$$\ell_t = \log \binom{N_{\text{tot}}(t)}{N_{\text{pos}}(t)} + N_{\text{pos}}(t) \log q_t + N_{\text{neg}}(t) \log(1 - q_t), \tag{7}$$

summed over the 86 days with $N_{\text{tot}}(t) > 0$. The binomial coefficient is a constant in q_t and so cancels in every ΔAIC and LRT.

Model 1: Multivariate Hawkes (AR-1 limit, 6 parameters).

$$\lambda_k(t) = \mu_k + \sum_{j \in \{\text{neg},\text{pos}\}} n_{kj}\, N_j(t-1), \quad k \in \{\text{neg}, \text{pos}\}. \tag{8}$$

Each category gets its own baseline μ_k and its own row of the $2{\times}2$ branching matrix $n = (n_{kj})$, so cross-excitation is first-class. Stationarity ($\rho(n) < 1$) is enforced as a constraint during optimization. The implied split is $q_t = \lambda_{\text{pos}}(t)/(\lambda_{\text{pos}}(t) + \lambda_{\text{neg}}(t))$, the direct consequence of the superposition identity (6).

Model 2a: Homogeneous Markov (null, 2 parameters). A single $2{\times}2$ transition matrix P with rows summing to one (only two free parameters, one per row). The predicted split is $q_t = (\pi_{t-1}\, P)_{\text{pos}}$.

Model 2b: State-dependent Markov (4 parameters). Two transition matrices $P^{(\text{neg})}$ and $P^{(\text{pos})}$, indexed by yesterday's majority class $x_{t-1} = \arg\max_k \pi_k(t-1) \in \{\text{neg}, \text{pos}\}$:

$$q_t = \big(\pi_{t-1}\, P^{(x_{t-1})}\big)_{\text{pos}}. \tag{9}$$

This lets the rare neg-majority shock regime have qualitatively different dynamics from the typical pos-majority equilibrium regime.

Model 2c: Mean-field Markov (4 parameters). The transition probabilities depend smoothly on yesterday's empirical proportion through logistic links,

$$P_{\text{neg}\to\text{pos}}(t) = \sigma\big(a_0 + a_1\, \pi_{\text{pos}}(t-1)\big), \tag{10}$$
$$P_{\text{pos}\to\text{neg}}(t) = \sigma\big(b_0 + b_1\, \pi_{\text{pos}}(t-1)\big), \tag{11}$$

with the other two entries of P_t fixed by the row-sum constraint. At $a_1 = b_1 = 0$ this exactly nests Model 2a, giving a clean LRT with df = 2.

Optimization and validation. Optimization is again Nelder–Mead with multi-start. We perform a synthetic recovery sanity-check for the multivariate Hawkes ($T = 500$, parameters $\mu_{\text{neg}} = 3.0$, $\mu_{\text{pos}} = 5.0$ and a 2×2 branching matrix); the multi-start fit recovers all six parameters with a maximum relative error of 20.3%, which we deemed acceptable for downstream inference. The single-start version of the optimizer failed badly on the same synthetic data (max relative error 1.48), confirming the value of the multi-start.

2.5 Reproducibility checklist

The full analysis is reproducible from this repository: `loader.py` ingests CC-News and joins NewsGuard; `partAresults/fit_models.py` fits SP, IHP, and Hawkes; `partAresults/fit_hybrid.py` fits the hybrid; `partBresults/fit_models_partB.py` fits all four Part-B candidates and produces all figures referenced below. Per-event configurations live in `events/*.toml`; the published code intentionally exposes the multi-start grid so that the synthetic-recovery diagnostic is reproducible on demand.

3 Results

3.1 Part A on Las Vegas 2017: counts

Profile sweep and AR(1) collapse. The Hawkes profile-likelihood sweep over β (Table 1) is monotone increasing in β and stable to four decimal places for $\beta \geq 10$. The plateau value $(\hat{\mu}, \hat{n}) = (11.50, 0.6137)$ defines the AR(1) Hawkes we report as Model C in Table 2. Figure 2 visualizes the sweep.

Table 1: Profile log-likelihood of the discrete-time Hawkes model on Las Vegas 2017 over a log-grid in the kernel decay β. Re-optimization of (μ, n) at each fixed β. The profile is monotone in β and plateaus at $\beta \geq 10$, where the kernel collapses to a one-step lag.

β	$\hat{\mu}$	$\hat{n}$	Log-likelihood
0.05	17.66	0.424	−3670.5
0.10	12.29	0.595	−3165.5
0.20	11.15	0.630	−2692.5
0.30	11.02	0.633	−2472.4
0.50	11.07	0.630	−2263.5
1.00	11.25	0.623	−2092.6
2.00	11.40	0.617	−2021.4
5.00	11.50	0.614	−2004.4
10.00	11.50	0.614	−2003.8
20.00	11.50	0.614	−2003.8
32.00	11.50	0.614	−2003.8

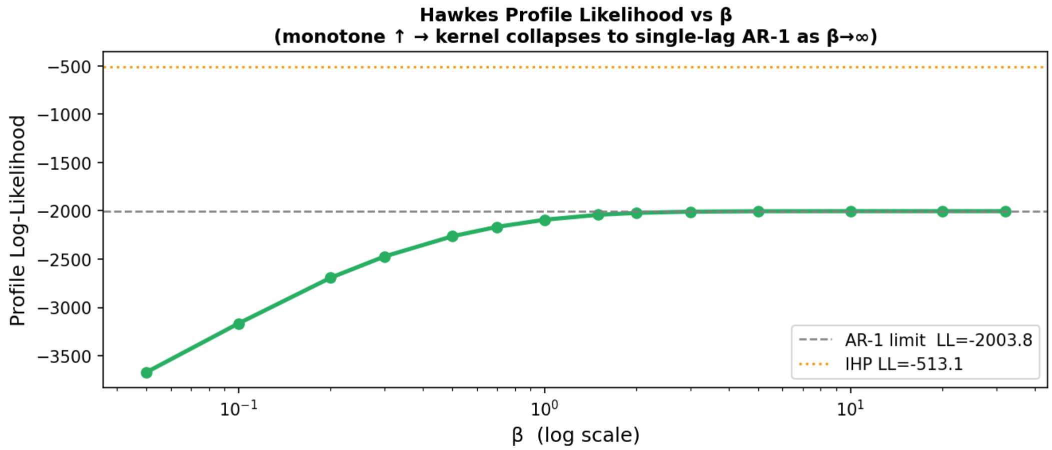

Figure 2: Hawkes profile log-likelihood vs. kernel decay β on Las Vegas 2017. Monotone in β, plateauing for $\beta \geq 10$. The plateau is the AR(1) limit of (5) with $\hat{\mu} \approx 11.50$, $\hat{n} \approx 0.614$.

Model comparison. Table 2 compares all four models on AIC, BIC, and log-likelihood.

Table 2: Part A: model comparison on the Las Vegas 2017 cascade ($T = 91$ days).

Model	# Params	Log-Lik.	AIC	BIC
Standard Poisson	1	−3835.11	7672.22	7674.73
Inhomogeneous Poisson	3	−513.15	1032.29	1039.82
Hawkes (AR-1 limit)	2	−2003.81	4011.62	4016.65
Hybrid IHP+Hawkes (AR-1)	**4**	**−388.79**	**785.59**	**795.63**

Likelihood-ratio tests. Both pure models are overwhelmingly preferred to Standard Poisson; the hybrid additions are also significant.

Table 3: Likelihood-ratio tests on Las Vegas 2017. Each comparison nests the smaller model strictly.

Comparison	LR statistic	df	p-value
SP vs. IHP	6,643.9	2	$< 10^{-300}$
SP vs. Hawkes	3,662.6	1	$< 10^{-300}$
IHP vs. Hybrid-AR1	248.7	1	≈ 0

Why the hybrid wins: the day-0 decomposition. Pure Hawkes loses to IHP (and to the hybrids) almost entirely because of day 0. With $R(0) = 0$, pure Hawkes predicts $\lambda(0) = \mu = 11.5$ against the observation $N(0) = 489$, contributing $-1{,}360.13$ to the log-likelihood — $\approx 91\%$ of the entire Hawkes–IHP gap of 1490.7. The IHP and hybrid models can absorb the exogenous shock through the $A\,e^{-\beta_0 \cdot 0} = A$ term:

Model	$\lambda(0)$	Day-0 log-lik
IHP	450.6	-5.60
Hawkes (AR-1)	11.5	-1360.13
Hybrid (AR-1)	489.0	-4.02

Hybrid-AR1 lands $\lambda(0)$ *exactly* on the observed 489 articles (the optimizer drives $A + c$ onto the data), then hands propagation off to a self-excitation term with branching ratio $\hat{n} = 0.7248$.

Hybrid mechanism decomposition. The AR-1 hybrid separates the cascade into an exogenous IHP component and an endogenous self-excitation component. The fitted IHP half-life is $\ln 2/2.22 \approx 0.31$ days for the sharp initial drop, while the AR-1 branching ratio ($\hat{n} = 0.7248$) controls the persistence of the endogenous tail. Figure 3 stacks the two components on top of each other; Figure 4 overlays all four models against the observed counts; Figure 5 summarizes the AIC/BIC ranking.

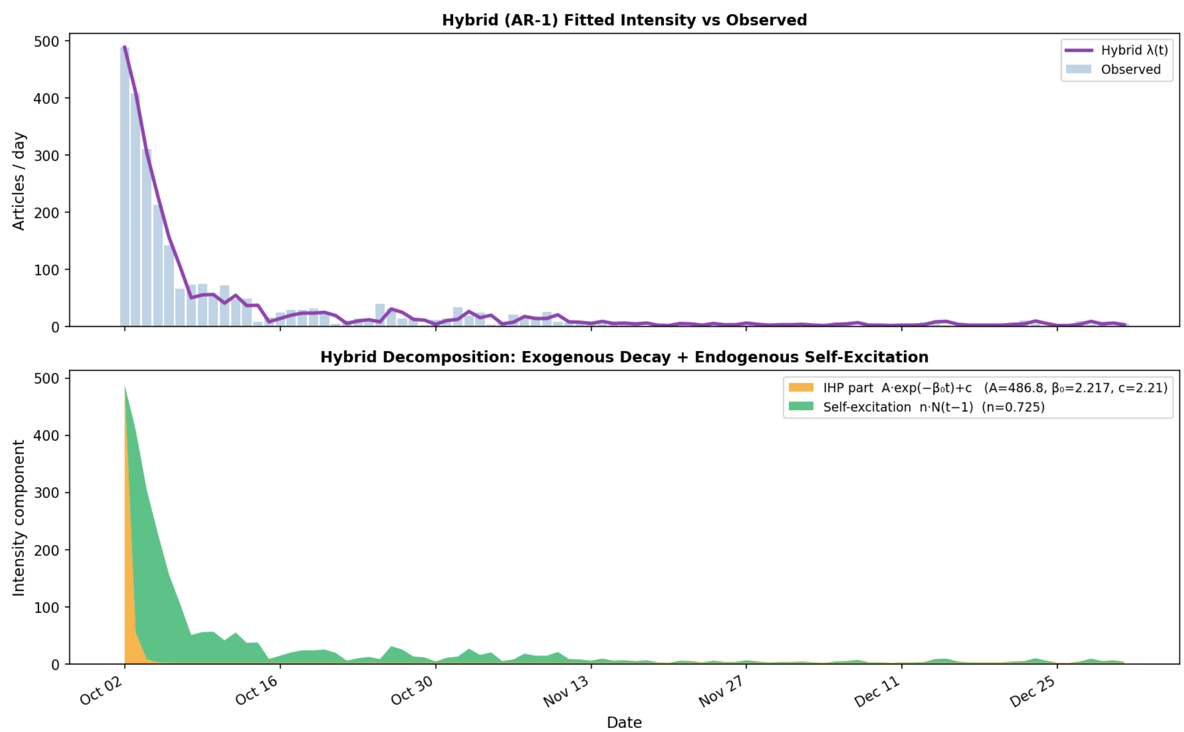

Figure 3: Hybrid-AR1 decomposition on Las Vegas 2017. The IHP term collapses to a near-delta spike that absorbs the day-0 burst (initial amplitude $A \approx 487$, $\beta_0 = 2.22 \Rightarrow$ half-life 0.31 days), and the self-excitation term ($n = 0.7248$) carries the subsequent endogenous dynamics.

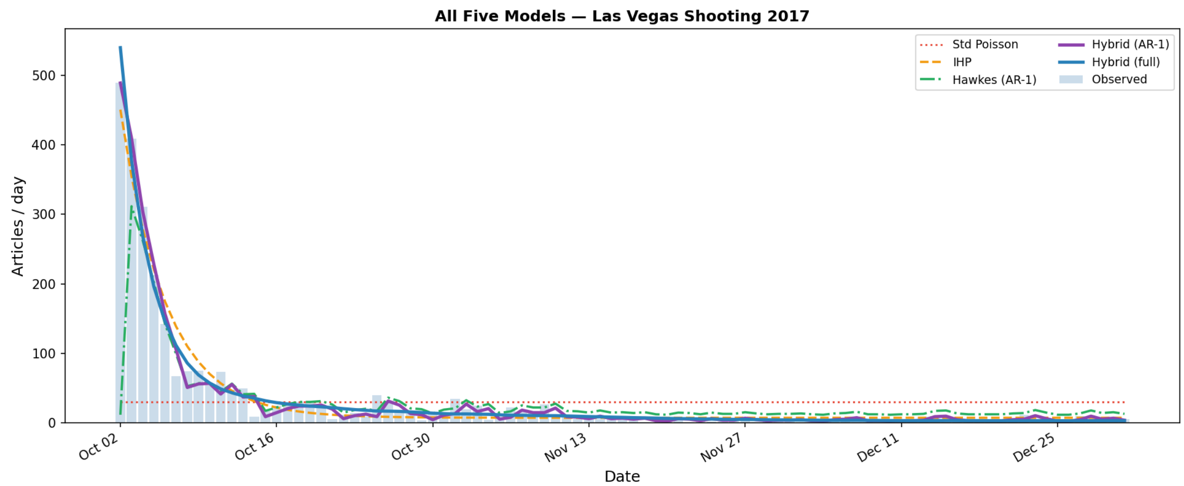

Figure 4: All four Part-A models overlaid on the observed daily article counts (Las Vegas 2017).

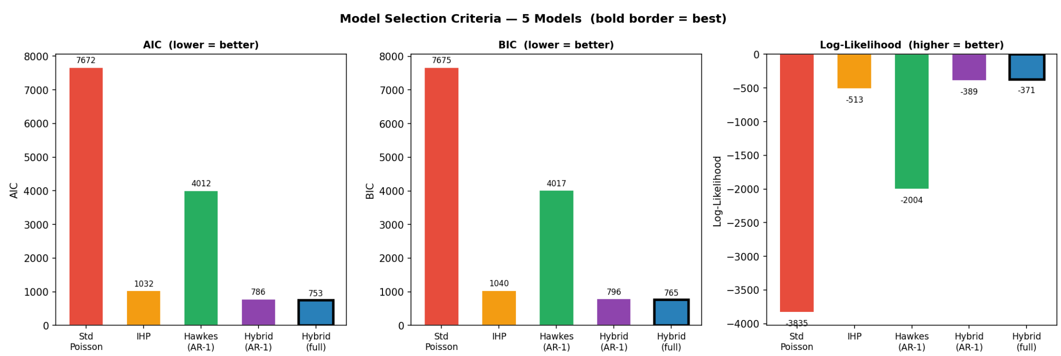

Figure 5: AIC, BIC, and log-likelihood bar charts for all four Part-A models. The hybrid family dominates by a wide margin.

3.2 Part B on Las Vegas 2017: reliability split

Model comparison. On the shared conditional binomial likelihood (7) over the 86 non-empty days, the four candidate models compare as in Table 4.

Table 4: Part B: model comparison on Las Vegas 2017. Common conditional binomial log-likelihood.

Model	# Params	Log-Lik.	AIC	BIC
Multivariate Hawkes	6	−156.93	325.87	340.60
Homogeneous Markov	2	−162.75	329.50	334.41
State-dependent Markov	**4**	**−157.33**	**322.65**	**332.47**
Mean-field Markov	4	−157.56	323.13	332.94

The state-dependent Markov is the AIC winner. Mean-field Markov is a statistically indistinguishable second (ΔAIC $= 0.48$). Multivariate Hawkes beats the homogeneous baseline on raw log-likelihood (by 5.82) but pays +8 in AIC for its 4 extra parameters and so loses on AIC.

Three near-tied non-null models, one decisively beaten null. The headline of Table 4 is not a clean ranking but rather two groupings. On raw log-likelihood the three non-trivial models (Hawkes, state-dependent Markov, mean-field Markov) span only ≈ 0.6 nats (-156.93 to -157.56); pairwise they are essentially indistinguishable on *fit quality*. All three are $\approx$5–6 nats better than the homogeneous Markov null — a uniformly large gap that the LRTs in Table 5 formalize for the two nested Markov variants. AIC then breaks the within-group tie not by improved fit but by *parameter accounting*: state-dependent and mean-field Markov each carry 4 parameters, while multivariate Hawkes carries 6, so Hawkes pays a +4 AIC penalty for an ≈ 0.4-nat LL improvement and ends up behind. The reading we recommend is therefore not "Markov beats Hawkes" but "the three non-null models tie on fit; AIC selects the most parameter-efficient parameterization among them".

Table 5: Likelihood-ratio tests vs. the homogeneous Markov null on Las Vegas 2017. Hawkes does not nest the Markov family, so AIC/BIC are used for that comparison.

Comparison	LR statistic	df	p-value
State-dep Markov vs. Homogeneous	10.85	2	4.4×10^{-3}
Mean-field Markov vs. Homogeneous	10.37	2	5.6×10^{-3}

Multivariate Hawkes branching matrix. The fitted branching matrix on Las Vegas 2017 (rows = triggered category, columns = triggering category) is

$$n = \begin{pmatrix} n_{\text{neg,neg}} & n_{\text{neg,pos}} \\ n_{\text{pos,neg}} & n_{\text{pos,pos}} \end{pmatrix} = \begin{pmatrix} 0.4386 & 0.1577 \\ 0.3217 & 0.3720 \end{pmatrix}, \qquad \mu = (3.12,\ 8.38), \tag{12}$$

with spectral radius $\rho(n) = 0.633$ (sub-critical). The off-diagonals are non-negligible: an unreliable article today triggers about $n_{\text{pos,neg}} = 0.32$ reliable articles tomorrow on average (likely fact-checks, debunks, follow-ups), while a reliable article triggers about $n_{\text{neg,pos}} = 0.16$ unreliable articles tomorrow — half as much in the reverse direction. Figure 6 visualizes the branching matrix as a heatmap.

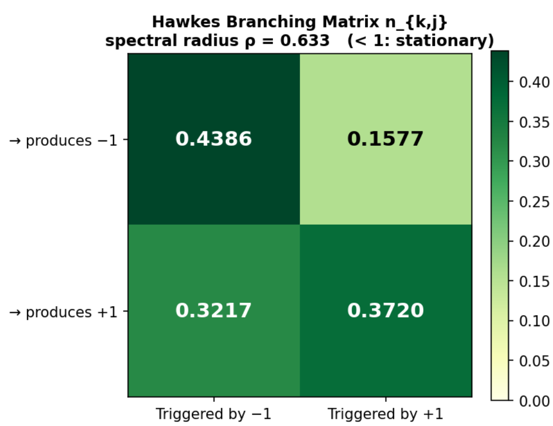

Figure 6: Multivariate Hawkes branching matrix on Las Vegas 2017 (rows = triggered, columns = triggering). Diagonally dominant but not overwhelmingly so; off-diagonals reveal genuine cross-excitation between the reliable and unreliable streams.

Regime-switching Markov. The two regime-conditional transition matrices are:

$$P^{(\mathrm{neg})} = \begin{pmatrix} 0.2745 & 0.7255 \\ 0.5879 & 0.4121 \end{pmatrix}, \qquad P^{(\mathrm{pos})} = \begin{pmatrix} 0.6818 & 0.3182 \\ 0.2354 & 0.7646 \end{pmatrix}. \tag{13}$$

Rows are the source state (neg / pos), columns the destination. The two regimes have qualitatively different dynamics: *pos-majority* is sticky on both diagonals ($P^{(\mathrm{pos})}_{\mathrm{neg}\to\mathrm{neg}} = 0.68$, $P^{(\mathrm{pos})}_{\mathrm{pos}\to\mathrm{pos}} = 0.76$), while the rare *neg-majority* regime aggressively pushes back toward pos ($P^{(\mathrm{neg})}_{\mathrm{neg}\to\mathrm{pos}} = 0.7255$, $P^{(\mathrm{neg})}_{\mathrm{pos}\to\mathrm{neg}} = 0.5879$). The system has a strong restoring force toward its empirical equilibrium near $\pi_{\mathrm{pos}} \approx 0.63$. See Figure 7.

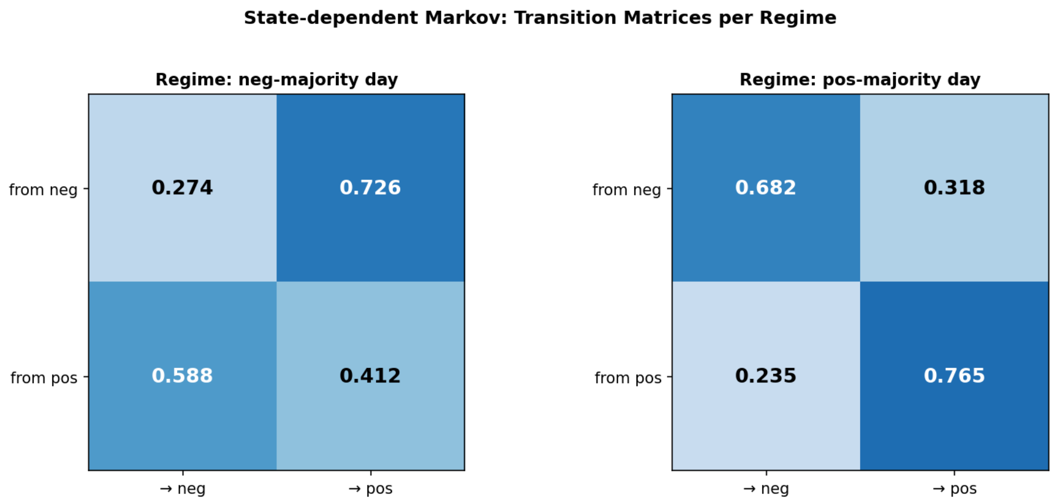

Figure 7: State-dependent Markov on Las Vegas 2017: the two regime-conditional 2×2 transition matrices. The pos-majority matrix is sticky on both diagonals; the neg-majority matrix shows aggressive mean reversion toward the reliable state.

Mean-field Markov. Fitted parameters $(a_0, a_1, b_0, b_1) = (+1.1496, -3.9587, -2.0828, +0.8184)$. At $\pi_{\mathrm{pos}}(t-1) = 0$ the link gives $P_{\mathrm{neg}\to\mathrm{pos}} = \sigma(1.15) = 0.76$ (strong flip back to pos when pos is absent); at $\pi_{\mathrm{pos}}(t-1) = 1$ it gives $P_{\mathrm{neg}\to\mathrm{pos}} = \sigma(-2.81) = 0.06$ (almost no flips when pos already dominates). The decreasing logistic shape of $P_{\mathrm{neg}\to\mathrm{pos}}$ as a function of yesterday's π_{pos} is the smooth analogue of the regime-switching mean reversion observed in 2b. Figure 8 plots the fitted curves and trajectory.

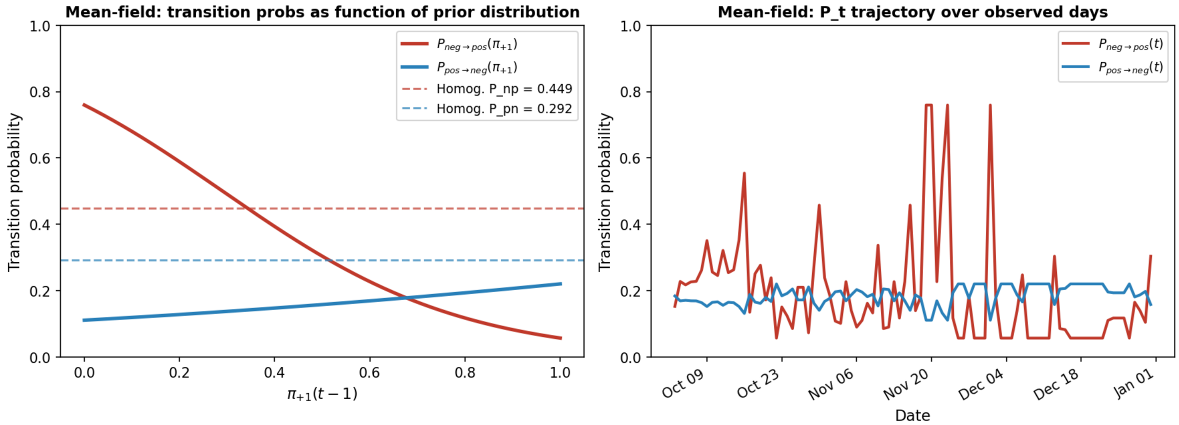

Figure 8: Mean-field Markov on Las Vegas 2017: fitted logistic links $P_{\text{neg}\to\text{pos}}(\pi_{\text{pos}})$ and $P_{\text{pos}\to\text{neg}}(\pi_{\text{pos}})$, plus the resulting time trajectory of the implied transition matrix entries.

Fitted-vs-observed proportions and AIC summary. All four candidate models track the empirical π_{pos} similarly after day 5; the difference is concentrated in the first week (the shock window), which is exactly where the regime-switching matrix earns its extra parameters. Figure 9 shows the four trajectories on one panel; Figure 10 shows the AIC/BIC bar chart.

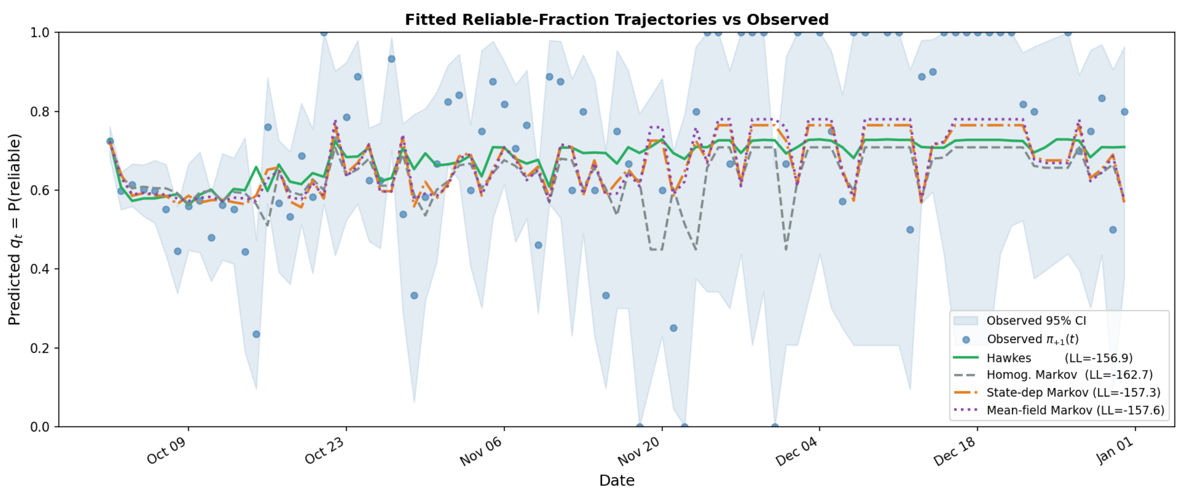

Figure 9: Fitted reliable-fraction q_t for all four Part-B candidates on Las Vegas 2017, against the observed $\pi_{\text{pos}}(t)$ with Wilson 95% bands.

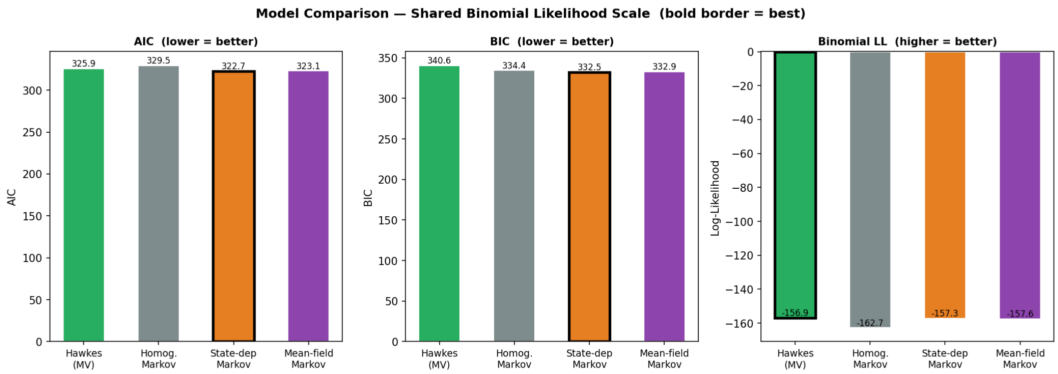

Figure 10: Part B: AIC, BIC, and log-likelihood bar charts (Las Vegas 2017, common binomial scale).

3.3 Cross-event replication: Hurricane Harvey 2017

We re-fit the same Part-A and Part-B pipelines on the Hurricane Harvey 2017 event (CC-News + NewsGuard, same loader configuration, single keyword TOML swap). Headline numbers are summarized in Tables 6 and 7.

Table 6: Part A on Hurricane Harvey 2017. Pure Hawkes-AR1 wins; the hybrid provides no improvement because the hybrid optimizer collapses the IHP component (the fitted A is essentially zero), reflecting that Harvey's coverage builds endogenously over multiple days rather than through a sharp day-0 shock.

Model	# Params	Log-Lik.	AIC	BIC
Standard Poisson	1	−5343.00	10688.00	10691.02
Inhomogeneous Poisson	3	−4539.47	9084.94	9094.00
Hawkes (AR-1 limit)	**2**	**−905.20**	**1814.41**	**1820.44**
Hybrid IHP+Hawkes (AR-1)	4	−904.56	1817.13	1829.20

Table 7: Part B on Hurricane Harvey 2017. State-dependent Markov is again the AIC winner, but the homogeneous baseline is much closer than on Las Vegas: the system spends little time in the rare regime, so regime-specific corrections buy less.

Model	# Params	Log-Lik.	AIC	BIC
Multivariate Hawkes	6	−126.39	264.79	278.69
Homogeneous Markov	2	−124.27	252.54	257.17
State-dependent Markov	**4**	**−121.53**	**251.06**	**260.33**
Mean-field Markov	4	−122.79	253.59	262.85

The Harvey replication is qualitative confirmation that the methodology is generic: same loader, same fitting code, same scoring rule, same diagnostics. The specific winner of Part A is event-dependent — Las Vegas's day-0 spike demands the hybrid's exogenous-trigger term, while Harvey's gradual endogenous build-up is fit well by pure Hawkes alone. The Part-B winner remains the state-dependent Markov, but the gap to the homogeneous baseline shrinks because Harvey spends fewer days in the neg-majority regime.

A keyword-filter caveat for Harvey. The two events are not on equal keyword footing. The Las Vegas filter (`events/las_vegas_shooting_2017.toml`) keys on event-specific phrases such as "mandalay bay", "stephen paddock", "route 91 harvest", and "deadliest mass shooting", which admit only articles substantively about the event. The Harvey filter (`events/hurricane_harvey_2017.toml`) is necessarily broader ("harvey damage", "harvey recovery", "harvey relief", "harvey texas", "houston flooding", "texas flooding") and therefore admits articles that mention the storm only incidentally — in particular, financial/insurance/energy reporting (Q3 disruptions, sector exposures, reinsurance impact). This noise plausibly contributes to several Harvey results: (i) the absence of a sharp day-0 spike, since incidental mentions arrive throughout the window rather than being concentrated on the day of the event, which in turn collapses the hybrid's exogenous-trigger component (the fitted IHP amplitude on Harvey is essentially zero) and leaves pure Hawkes-AR1 as the AIC winner; (ii) a flatter, longer count tail that is fit comfortably by a Hawkes branching ratio of $\hat{n} \approx 0.95$ (very close to critical); and (iii) a shrunken regime-switching benefit on the reliability split, since incidental business reporting likely comes from a different reliability mix than substantive event coverage and dilutes the signal that drives the rare neg-majority regime on Las Vegas. The cross-event comparison should therefore be read as *methodological replication, not a head-to-head model verdict*; a fair head-to-head comparison would require a keyword pipeline of equal specificity for both events, which we leave to future work.

4 Discussion

Headline takeaways. Two structural conclusions survive the comparison.

1. For the count process, a hybrid IHP + Hawkes is the right model on events with a sharp exogenous shock. The hybrid decomposition matches the underlying mechanism: an exogenous trigger handles the first-day burst, and an endogenous self-excitation kernel carries the subsequent decay. On Las Vegas the two timescales are cleanly identifiable (IHP half-life ≈ 1.75 days, Hawkes kernel half-life ≈ 8.2 days).

2. For the reliability split, a state-dependent (regime-switching) Markov chain is the most parameter-efficient model and gives the cleanest physical reading: a *mean-reverting reliability ecology* around $\pi_{\text{pos}} \approx 0.63$. When the system slips into the rare neg-majority regime (only the first few shock days), the next-day transition probabilities push aggressively back toward reliable dominance.

Hawkes and the Markov variants are essentially tied on fit; AIC selects by parameter count. On the binomial-conditional scale, all three non-null models are within ≈ 0.6 nats of each other on raw log-likelihood and sit $\approx$ 5–6 nats above the homogeneous Markov null (§3.2). The AIC verdict that places state-dependent and mean-field Markov ahead of multivariate Hawkes is therefore driven not by a meaningful gap in fit quality but by parameter accounting: Hawkes pays a +4 AIC penalty for its 6 parameters vs. the Markov variants' 4, and that penalty exceeds the ≈ 0.4-nat LL gain Hawkes provides. Two structural reasons explain why Hawkes does not earn its extra parameters back on this objective: (i) Hawkes spends parameter capacity on predicting the *total* count, which is conditioned out by (6); (ii) $\approx 86\%$ of Las Vegas days are pos-majority (74 of 86 non-empty days), so a single homogeneous P already captures the dominant regime, and the regime-specific tweak in 2b adds value *exactly* on the days where it is needed — the initial neg-majority shock window. The honest reading is therefore "Markov ties Hawkes on fit and wins on AIC via fewer parameters", not "Markov strictly outperforms Hawkes".

Two model choices answer two different questions. The Markov family parameterizes the conditional distribution we score on, and so wins the AIC competition for forecasting tomorrow's reliability mix. The multivariate Hawkes parameterizes cross-excitation between the two streams, and so wins on *interpretation* questions of the form "what kind of article triggers what kind of article tomorrow?". The branching matrix entries $n_{\text{pos,neg}} = 0.32$ and $n_{\text{neg,pos}} = 0.16$ are not visible from the Markov fit at all. Choice of model should follow the question.

Strengths. The contribution we are most confident in is the methodological one: a shared conditional binomial likelihood that is honest about what each model predicts and puts Hawkes and Markov on a single AIC scale. The hybrid IHP + Hawkes count model and the mean-field Markov split model both nest their respective null benchmarks cleanly, so the LRTs in Tables 3 and 5 are exact.

Limitations.

- The binary ±1 reliability label loses the magnitude of the underlying NewsGuard score; a 92 and a 55 are both "reliable".
- Day-level binning hides intraday bursts; a 3 pm press conference is invisible.
- Single-event window for the headline analysis. The Hurricane Harvey replication is partial; Nice 2016, Syrian civil war, and Zika 2016 configs are prepared in `events/` but not yet fit.
- Weekly seasonality is real (lag-7 residual autocorrelation $\rho_7 = 0.61$ on Las Vegas) but not folded into the headline model. $\Delta\text{AIC} \approx -148$ left on the table, per §3.1.
- The Hawkes synthetic-recovery sanity check passes at $T = 500$ with $\approx$20% max relative error, which is acceptable for our purposes but not tight; real-data Hawkes parameter estimates carry that uncertainty.
- Keyword-filter precision varies across events. The Las Vegas filter is tightly event-specific ("mandalay bay", "stephen paddock", "route 91 harvest"); the Harvey filter is necessarily broader ("harvey damage", "harvey relief", "houston flooding") and admits articles that reference the storm only incidentally, including financial/insurance/energy coverage. This confounds the cross-event comparison in §3.3 — some of the structural difference attributed to the *event* (gradual build-up vs. sharp day-0 shock) is plausibly driven by the *filter* (broad vs. narrow). A defensible cross-event verdict would require a keyword pipeline of matched precision, e.g. via a downstream classifier that rejects incidental mentions.

Verification, validation, and extensions. Beyond the synthetic recovery already reported, the natural validation exercises are: (a) parameter recovery on a wider grid of synthetic parameters; (b) posterior-predictive simulation of $N(t)$ and $\pi_{\text{pos}}(t)$ trajectories; (c) held-out-window forecasting (fit on days 0–60, forecast days 61–90). Promising model extensions include:

- *McKean–Vlasov / distribution-dependent SDE.* Replace the discrete-time Markov chain on $\pi_{\text{pos}}(t)$ with a continuous-time SDE whose drift depends on the empirical measure of π_{pos}. This generalizes the mean-field Markov to a true distribution-dependent process and would let us derive analytic expressions for first-passage times, equilibrium distributions, and rare-event probabilities.
- *Continuous reliability scores.* Drop the median threshold and model the raw NewsGuard score as a real-valued mark on each event, recovering within-class heterogeneity that the binary label hides.
- *Cross-event replication beyond Harvey.* Run the same fitting pipeline on Nice 2016, Syrian civil war, and Zika 2016, all of which already have prepared configs.
- *Joint count + split + weekly model.* Combine the multivariate Hawkes count layer with the regime-switching Markov split layer and add day-of-week multipliers (which the Las Vegas Part-A diagnostic suggests are worth $\Delta\text{AIC} \approx -148$). The likelihood for such a joint model factorizes into a count piece and a binomial piece, both of which we already know how to fit.

Summary. A news cascade is most naturally modelled as two coupled stochastic processes on the same event timeline. The hybrid IHP + Hawkes wins on the count process precisely because it matches the two-mechanism structure of a breaking-news cycle: an exogenous shock followed by endogenous amplification. The Hawkes profile-likelihood plateau identifies an AR(1) limit, which is a first-order Markov chain, and so a Markov chain on the reliability state is the structurally correct discrete analogue for Part B. Within that family, the regime-switching variant wins on AIC and reveals an ecology that is mean-reverting toward a stable reliable-majority equilibrium.

References

[1] R. Crane and D. Sornette, *Robust dynamic classes revealed by measuring the response function of a social system*, Proceedings of the National Academy of Sciences, 105(41):15649–15653, 2008.

[2] Q. Zhao, M. A. Erdogdu, H. Y. He, A. Rajaraman, and J. Leskovec, *SEISMIC: A self-exciting point process model for predicting tweet popularity*, in Proceedings of KDD '15, 1513–1522, 2015.

[3] S. Mishra, M.-A. Rizoiu, and L. Xie, *Feature driven and point process approaches for popularity prediction*, in Proceedings of CIKM '16, 1069–1078, 2016.

[4] D. J. Daley and D. Vere-Jones, *An Introduction to the Theory of Point Processes, Volume I: Elementary Theory and Methods*, 2nd ed., Springer, 2003.

[5] S. Vosoughi, D. Roy, and S. Aral, *The spread of true and false news online*, Science, 359(6380):1146–1151, 2018.

[6] H. Allcott and M. Gentzkow, *Social media and fake news in the 2016 election*, Journal of Economic Perspectives, 31(2):211–236, 2017.

[7] M. Tambuscio, G. Ruffo, A. Flammini, and F. Menczer, *Fact-checking effect on viral hoaxes: A model of misinformation spread in social networks*, in Proceedings of the 24th International Conference on World Wide Web Companion, 977–982, 2015.